

Daily Log Project M.Sc. ECMM454

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Project Title: Multi-Objective Optimisation of Swansea Bay Tidal Lagoon Operations Using NSGA-II

Internal Supervisor: Dr. Weliang Zang

External Supervisor: None

Module: ECMM454 – MSc Computer Science Business Project

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Student AI Declaration

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☒ I have used GenAI tools to:

- ☒ develop ideas.
- ☐ assist with research or gathering information.
- ☐ help me understand key theories and concepts.
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Daily Log of Activity

Date	Activity	Outcomes	Comments
Week 1-2: Project Initiation and Proposal Development			
01/04/2025 – 05/04/2025	Project proposal literature review initiation and methodology framework establishment Technical Activities: - Conducted comprehensive literature review of NSGA-II applications in renewable energy optimisation, particularly in marine energy - Analysed Xue et al. (2020) baseline methodologies for tidal range structures (TRS) optimisation - Reviewed British Oceanographic Data Centre (BODC) tidal datasets for Swansea Bay, particularly for Mumbles station - Established project scope: 0-D modelling framework with NSGA-II multi-objective optimisation	Identified critical research gap: Existing literature focuses predominantly on single-objective energy maximisation rather than multi-objective energy-cost trade-off analysis Confirmed availability of 2011-2012 BODC tidal height datasets for model variation Established methodological foundation combining 0-D modelling with evolutionary multi-objective optimisation	Technical Decisions: - Selected NSGA-II over NSGA-III due to bi-objective nature of the problem (energy output vs. unit cost) - Chose Java implementation for computational efficiency and object-oriented architecture scalability - Adopted Swansea Bay case study for practical relevance and data availability Progress Reflection: The literature review revealed a significant methodological gap in existing tidal lagoon optimisation research. While previous studies by Xue et al. and others demonstrate effective energy maximisation through genetic algorithms, they fail to address the economic viability constraints that led to Swansea Bay's rejection. This insight fundamentally shaped the research direction towards multi-objective optimisation, recognising that operational strategies must balance energy generation with cost-effectiveness for commercial viability.
			Challenges Encountered: Limited NSGA-II applications specifically for tidal energy systems required broader scope covering hydropower and marine renewable energy fields

			- Initial concern regarding computational complexity of 0-D model integration with population-based optimisation Mitigation Strategies: - Extended literature review to include related renewable energy optimisation studies - Planned modular architecture development to enable efficient computational scaling
05/04/2025 – 10/04/2025	SMART objectives formulation and project planning methodology Technical Activities: - Develop Specific, Measurable, Achievable, Relevant, Time-Based (SMART) project objectives - Established five-phase project timeline: Literature Review -> Model Development -> Framework Implementation -> Experimental Analysis -> Reporting - Defined primary deliverables: 0-D simulation framework, NSGA-II optimisation algorithm, benchmarking analysis, academic report	Completed SMART objectives framework ensuring project aligns with MSc timeline constraints (link: Work plan - Emre.xlsx) Established risk assessment and mitigation strategies for each development phase Defined success criteria: Energy output validation within 5% of published literature, successful Pareto front generation	Methodological Decisions: - Selected 0-D modelling approach over 2-D for computational efficiency while maintaining accuracy - Established benchmarking strategy against Xue et al. and other existing literatures - Planned systematic parameter tuning methodology for algorithm optimisation Progress Reflection: The SMART framework provided essential structure for managing the project's technical complexity. The decision to focus on 0-D modelling rather than more computationally intensive 2-D approach was crucial for enabling extensive optimisation experiments within the available timeframe. This trade-off between model fidelity and computational tractability reflects a key consideration in applied optimisation research.
10/04/2025 – 15/04/2025	Proposal structure development and technical specification refinement Technical Activities:	Completed draft proposal ready for supervisor review Established technical foundation for implementation phase	Technical Architecture Planning: Simulation Layer: TidalSimulator, Individual, Lagoon configuration classes

	• Drafted comprehensive project proposal including background, methodology, and implementation plan • Specified technical requirements: Java development environment, BODC data processing capabilities, statistical analysis tools • Developed initial class architecture for modular 0-D simulation framework • Established evaluation criteria and performance metrics for algorithm validation	• Defined clear interfaces between simulation and optimisation components	• Optimisation Layer: NSGA-II implementation, genetic operators, population management • Analysis Layer: Convergence Tracking, statistical validation, benchmarking tools Progress Reflection: The proposal development phase revealed the importance of modular architecture design. The separation of concerns between simulation, optimisation, and analysis components would prove crucial for systematic development and debugging in subsequent phases. Quality Assurance: • Internal review conducted for technical accuracy and feasibility assessment • Cross-referenced methodology with established NSGA-II implementation guidelines • Validated proposal scope against module learning outcomes and assessment criteria
Week 3-4: Supervisor Consultation and Proposal Finalisation			
15/04/2025	Supervisor meeting and proposal review session	• Supervisor confirmed appropriateness of 0-D modelling approach for optimisation research • Validation of NSGA-II selection for multi-objective energy-cost optimisation problem • Approval of Swansea Bay case study scope and BODC dataset utilisation • Clarification of academic contribution expectations and novel methodology requirements	Technical Guidance Received: • Recommendation to implement comprehensive parameter tuning methodology • Emphasis on statistical validation across multiple datasets for robustness assessment • Suggestion to explore operational complexity modelling in objective function design Key Questions Resolved: 1. Methodology Scope: Confirmed focus on operational optimisation rather than structural design parameters

			2. Validation Strategy: Approved benchmarking against existing literature 3. Data Requirements: Validated sufficiency of 2011-2012 BODC datasets for initial model development 4. Timeline Feasibility: Confirmed realistic scope for MSc project duration with appropriate risk mitigation Progress Reflection: Supervisor consultation provided essential validation of the technical approach and highlighted areas requiring particular attention for the proposal document. The emphasis on statistical rigour and multi-dataset validation would significantly influence the experimental design in later phases. The discussion clarified the balance between technical innovation and practical implementation within academic constraints. Action Items Generated: • Implement systematic parameter tuning framework • Develop multi-dataset validation methodology • Enhance objective function design for realistic operational complexity modelling • Establish comprehensive statistical analysis protocols
16/04/2025 – 29/04/2025	Proposal Revision and IEEE referencing implementation Technical Activities: • Implemented supervisor feedback: transition to IEEE numerical	• Successfully finalised proposal to be submitted via ELE platform by 2nd of May • Achieved full compliance with ECMM454 submission requirements • Established foundation for implementation phase initiation	Research Ethics Considerations: • Proper attribution of AI assistance in proposal development process • Transparent documentation of ChatGPT utilisation for literature synthesis and formatting

	referencing system - Updated institutional branding and formatting standards compliance - Enhanced references section with additional peer-reviewed literature - Documented AI tool usage (ChatGPT) for academic integrity compliance Quality Assurance Measures: - Systematic checklist verification of all supervisor-requested revisions - Cross-verification of IEEE referencing format against university guidelines - Final formatting review and academic presentation standards		Adherence to university guidelines for ethical AI tool acknowledgement Progress Reflection: The revision process reinforced the importance of academic rigour and proper attribution in research documentation. The systematic approach to incorporating supervisor feedback demonstrated professional responsiveness and attention to detail essential for successful academic research.
02/05/2025	Deadline for Project Proposal		
03/05/2025 – 13/05/2025	Exam Period		
Week 5-6: Core Implementation Phase			
14/05/2025 – 27/05/2025	Java framework development Technical Milestone Achieved: Successfully developed core Java classes for 0-D tidal lagoon simulation framework with	Core Java Classes Implemented: Lagoon.java: Static configuration class containing all fixed parameters for Swansea Bay Tidal Lagoon with utility methods for cost calculations	Technical Issues Identified and Resolved: Pareto Front Generation Problem: Initial testing revealed configuration-dependent behaviour with identical objective values across multiple solutions Temporal Scaling Errors: Annual vs. daily/weekly

	modular architecture supporting NSGA-II integration	• Individual.java: Solution representation class encoding operational strategies as [Hs, He] control parameter pairs per half-tide, with NSGA-II metadata (rank, crowding distance) and objective values (energy output, unit cost) • TideDataReader.java: Data processing utility for reading and parsing tidal height data from BODC text files into simulation ready format. • Main.java: Program entry point orchestrating workflow. Loads tidal data, generates test individuals, executes simulation, and displays energy output results for validation • IndividualGenerator.java: Factory class for creating randomly initialised individuals with feasible Hs and He control parameters within physical constraints (0.5-4.0m) for population-based optimisation • TidalSimulator.java: Core 0-D simulation engine implementing mass balance calculations, turbine operation logic, and energy generation computation based on head differences and flow dynamics • ObjectiveFunction.java: Multi-objective evaluation class calculating conflicting objectives. Maximising energy output (MWh) and minimising unit cost (£/MWh based on £1.3B capital cost), with robust handling of infeasible solutions	configurations produced inconsistent energy outputs (50 GWh vs. expected 400-500 GWh range) • Objective Function Correlation: Perfect correlation between energy and cost objectives prevented meaningful trade-off analysis Diagnostic Analysis: • Annual configuration yielding single Pareto front suggested insufficient solution diversity • Multiple identical solutions indicated potential convergence issues or inadequate exploration • Energy output discrepancies suggested systematic errors in temporal scaling factors Progress Reflection: The implementation phase revealed that the complexity of translating theoretical optimisation frameworks into practical simulation tools. The emergence of Pareto front generation issues highlighted the critical importance of objective function design in multi-objective optimisation. These early challenges provided valuable insights into the nuanced relationship between simulation parameters and optimisation performance. Supervisor Consultation Results: • Confirmed acceptability of 2011-2012 dataset for validation • Approved Random Search as baseline hyperparameter tuning method • Identified key configuration parameters required systematic tuning • Established debugging strategy for Pareto front resolution
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			Next Phase Preparation: • Priority on resolving objective function correlation issues • Investigation of temporal scaling factors in energy calculations • Validation of simulation parameters against published literature benchmarks • Development of solution diversity enhancement strategies
Week 7-8: Algorithm Implementation and Issue Resolution			
28/05/2025 – 10/06/2025	Complete NSGA-II implementation and objective function breakthrough Major Technical Breakthrough Achieved: Successfully resolved Pareto front generation issues through comprehensive objective function redesign, enabling realistic multi-objective trade-off analysis	NSGA-II Implementation Components: • Population.java: Statistics tracking and sorting algorithms, utility methods for generation-based operations, memory-efficient population handling for large-scale experiments • ParetoDominance.java: Implements efficient dominance comparison logic, $O(MN^2)$ total complexity for M objectives and N solutions, robust handling of constraint violations and infeasible solutions • CrowdingDistance.java: Maintains solution spread across Pareto front, prevents premature convergence to local optima, implements boundary point protection mechanisms • GeneticOperators.java: Implements specialised crossover (SBX, UNIFORM, HALFTIDE) and mutation (POLYNOMIAL, GAUSSIAN, OPERATIONAL) operators for tidal lagoon operational parameter evolution	Critical Technical Resolution (Objective Function Redesign): Problem Identified: Artificial correlation between energy output and unit cost prevented meaningful Pareto front generation Solution Implemented: Introduction of operational complexity modelling: Cost = baseCapitalCost + operationalPenalty (avgHead^{2.5}) + energyPenalty(energy^{1.8}) Technical Improvements: • Time Scaling Corrections: Updated half-tide duration from 12 to 6.12 hours based on literature validation • Penalty Term Calibration: Balanced capital cost dominance through operational complexity factors • Enhanced Physics Simulation: Corrected energy calculations and fixed time step computations Performance Results:

			• Configuration Scaling Analysis: Daily: 510 GWh (optimal day * 365) Weekly: 490 GWh (optimal week * 13) Annual: 350 GWh (most realistic, no scaling) • Literature Validation Challenge: Annual output below published range (400-720 GWh) • Stratified Sampling Solution: Implemented seasonal approach improving output from 350 GWh to 410 GWh (inspired by Xue et al.) Advanced Optimisation Enhancements: • Smart Population Initialisation: Energy-focused bias using higher head values • Genetic Operator Optimisation: Comparative analysis yielding 440 GWh final output • Multi-Component Cost Structure: Realistic LCOE model replacing artificial penalties Progress Reflection: This phase represented a critical turning point in the research, transforming a technically flawed implementation into a robust multi-objective optimisation framework. The objective function redesign process provided deep insights into the relationship between operational strategies and economic viability in tidal lagoon systems. The breakthrough demonstrated the importance of iterative refinement and the value of systematic debugging in complex optimisation research. Research Contribution Insights:
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			• Multi-Objective Advantage: Literature focus on single-objective energy maximisation vs. comprehensive energy-cost trade-off analysis • Methodological Gap Identification: Introduction of multi-objective framework addresses literature limitation in economic feasibility assessment • Industry Alignment: Cost benchmarking achieved £598-665/MWh range with realistic operational complexity modelling
Week 9-10: Literature-Based Validation and Physics Enhancement			
11/06/2025 – 24/06/2025	Industry Standard cost modelling implementation and turbine physics enhancement Major Methodological Advancement: Replaced artificial penalty-based cost function with realistic industry-standard LCOE model using actual Swansea Bay Contract for Difference (CfD) submission data.	Literature-Based Cost Function Implementation: Technical Foundation: Segura et al. (2017) LCEO methodology combined with Real-world capital cost structure based on CfD documentation • 35-year operational lifetime (annualised CAPEX * 35) • 6% Weighted Average Cost of Capital (WACC) • Detailed Operations & Maintenance (O&M) costs from 2015 industry submission Enhanced TidalSimulator.java Implementation: • Industry-Aligned Turbine Efficiency Curves: Replaced Binary on/off control with graduated operation • Hysteresis Effects Modelling: Realistic turbine start-up and shut-down behaviour • Hill Chart Efficiency Integration: Andritz Hydro turbine specifications implementation • Operational Threshold Validation: Updated	Critical Technical Breakthrough – Cost Modelling: Problem Identified: Artificial penalty factors lacked industry validation and created unrealistic cost projections Root Cause Analysis: • Capital cost dominance: Operational strategies influence only 5-10% variable portion • Lack of empirical basis for penalty factors (HEAD_VARIATION_COST, ENVIRONMENTAL_COST) • Disconnect between academic modelling and industry financial realities Solution Implementation: Realistic LCOE Components (C1-C6 Structure): • C1: Capital expenditure amortisation • C2: Fixed operational costs • C3: Variable maintenance costs • C4: Environmental compliance costs • C5: Grid connection charges

		minimum operating head from 0.5m to 1.0-2.0 based on literature	• C6: Insurance and regulatory costs Industry Validation Achievements: • Cost reduced from artificial £598-665/MWh to realistic £250-270/MWh • Alignment with £174-300/MWh benchmarks • Validation against Petley & Aggidis (2016) industry standards Physics Enhancement Results: • Separated Efficiency Modelling: Hydraulic (head-dependent) and generator (97%) efficiencies • Graduated Control Logic: Replaced binary operation with continuous efficiency curves • Realistic Operating Constraints: Industry-validated head thresholds and turbine characteristics Progress Reflection: This phase represented a fundamental paradigm shift from academic modelling to industry-aligned simulation. The recognition that artificial penalties lack empirical validation led to complete cost model reconstruction using authentic industry data. This breakthrough provided crucial insights into the economic realities of tidal lagoon operations, explaining why existing literature focuses on energy maximisation rather than cost optimisation. Research Contribution: • Industry-Standard Alignment: Transition from artificial academic modelling to real-world financial frameworks • Methodological Advancement:
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			Demonstrated importance of empirical validation over theoretical penalty systems • Economic Reality Integration: Bridged gap between academic optimisation and commercial viability assessment
Week 11-14: Systematic Parameter Tuning and Statistical Validation			
25/06/2025 – 22/07/2025	Comprehensive parameter tuning methodology and multi-dataset validation Major Milestone: Implementation of systematic two-phase parameter optimisation with statistical validation across multiple datasets.	Dataset Validation Infrastructure: • DatasetComparator.java: Comprehensive comparison tool for 2011/2012 datasets • Baseline Performance Analysis: Confirmed <10% difference in tidal characteristics • Statistical Metrics: Mean difference 1.3%, Range difference 14.1%, Standard deviation difference 2.6% • Grid Search Validation: <1.2% performance difference between datasets confirming dataset suitability Phase 1: Operator Selection (9 Configurations Tested) Methodology: • Systematic evaluations of crossover/mutation operator combinations • Statistical analysis with 5 independent runs per configuration per dataset • Performance metrics: energy output, cost efficiency, robustness assessment Results: • Winner: HALFTIDE_GAUSSIAN combination • Performance: 430.44 GWh vs. 424.3 GWh for UNIFORM_GAUSSIAN • Cost Efficiency: £245.8/MWh (2011) and £264.2/MWh (2012)	Methodological Decision – Random Search Replacement: Replaced originally planned Random Search baseline with systematic manual tuning for superior parameter space exploration and enhanced scientific rigour. Performance Validation: • Robustness Assessment: <15% performance difference across datasets for algorithm stability • Statistical Significance: 5 independent runs per configuration for confidence interval establishment • Reproducibility: Automated testing and result archiving for scientific verification Progress Verification: The systematic parameter tuning phase established the methodological rigour essential for credible research validation. The identification of optimal NSGA-II configuration through comprehensive statistical analysis provided confidence in subsequent benchmarking results. The decision to replace Random Search with systematic manual tuning reflected a commitment to thorough parameter space exploration over convenience sampling. Research Methodology Achievement: • Evidence-Based Parameter Selection: Optimal configuration

		• Robustness: 6.1% energy difference between datasets (excellent <15% threshold) Technical Insight: HALFTIDE crossover preserves operational relationship while Gaussian mutation provides effective local search capabilities Phase 2: Parameter Optimisation Population Size Analysis: • Winner: 400 Individuals • Rationale: Optimal exploration-exploitation balance (436.22 GWh performance) • Computational Efficiency: 19.02 vs. 58.48s for 800 population size • Scalability Consideration: Maintained solution quality without excessive overhead Probability Parameter Tuning: • Crossover Probability: 0.9 (Standard NSGA-II value with robust cross-dataset performance) • Mutation Probability: 0.015 (achieved highest energy 439.38 GWh with £242.2/MWh cost) • Tournament Size: 2 (best energy performance 439.86 GWh with fastest convergence) ManualTuningRunner.java implementation: • Comprehensive testing infrastructure with automated results formatting • Excel-compatible output for systematic analysis and tracking • 250+ optimisation runs with comprehensive performance documentation	identification through systematic experimentation • Multi-Dataset Validation: Ensures real-world applicability across varying tidal conditions • Statistical Rigour: Comprehensive framework for reproducible optimisation research • Industry-Standard Approach: Follows established NSGA-II optimisation research methodologies Optimal Configuration Confirmed: • Algorithm: NSGA-II with HALFTIDE_GAUSSIAN operators • Parameters: Population=400, Pc=0.9, Pm=0.015, Tournament=2 • Performance: 442.50 GWh at £242/MWh (2011), 410.90 GWh at £259/MWh (2012) • Validation: <15% dataset variation confirming algorithm robustness
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Week 15-18: Final Documentation and Project Completion			
23/07/2025 – 14/08/2025	Comprehensive project documentation and deliverable completion	Project Deliverables Successfully Completed: 1. Video Presentation (Submitted: 8 August 2025) 2. Project Logbook Completion (Submitted: 15 August 2025) 3. Final Report Development (Submitted: 15 August 2025)	Critical Research Insights Documented: Economic Optimisation Reality: • Infrastructure vs. Operational: Real cost optimisation occurs at infrastructure level (Cardiff link: 17.2% reduction) rather than algorithmic improvements • Policy Implications: Tidal lagoons require massive government intervention rather than operational optimisation for viability • Technology Assessment: Realistic baseline for tidal energy sector expectations vs. optimistic academic projections Methodological Contributions: • Multi-Objective Framework: Addresses literature limitation in economic feasibility assessment • Industry-Standard Integration: Transition from artificial academic penalties to real-world financial models • Statistical Validation: Comprehensive parameter tuning methodology for reproducible optimisation research Final Performance Results: • Energy Output: 442.50 GWh (2011), 410.90 GWh (2012) • Cost Efficiency: £242/MWh (2011), £259/MWh (2012) • Literature Alignment: Results within published range while providing superior cost-efficiency insights • Algorithm Robustness: <15% performance variation across datasets

			Professional Development Outcomes: • Technical Competence: Advanced multi-objective optimisation implementation • Research Methodology: Systematic experimental design and statistical validation • Industry Integration: Real-world parameter validation and economic modelling • Academic Communication: Professional documentation and presentation skills Progress Reflection: The project completion phase synthesised four months of intensive research and development into comprehensive academic documentation. The integration of technical achievements with critical analysis of industry implications demonstrated the project's contribution to both academic knowledge and practical renewable energy policy. The systematic documentation approach ensured reproducibility and provided a foundation for future tidal energy optimisation research. Research Impact Assessment: Successfully bridged the gap between academic energy optimisation and commercial viability reality, providing transparent economic modelling and realistic expectations for future tidal lagoon projects. The research validates multi-objective approaches while acknowledging fundamental economic constraints that limit operational optimisation impact.
15/08/2025	FINAL SUBMISSION		